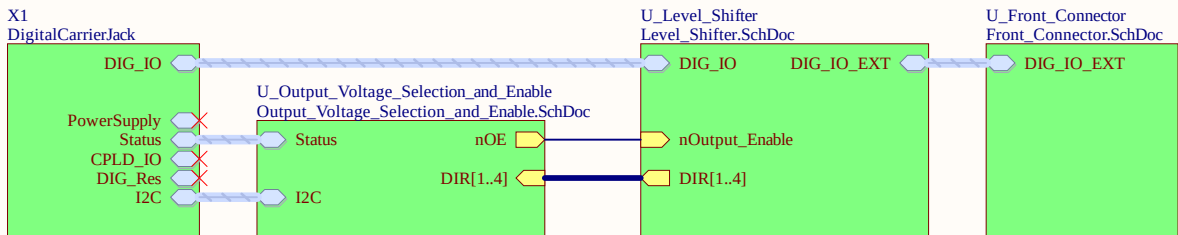
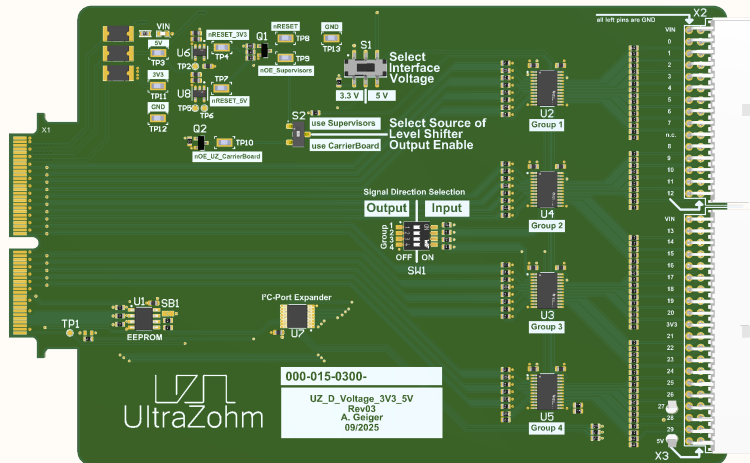
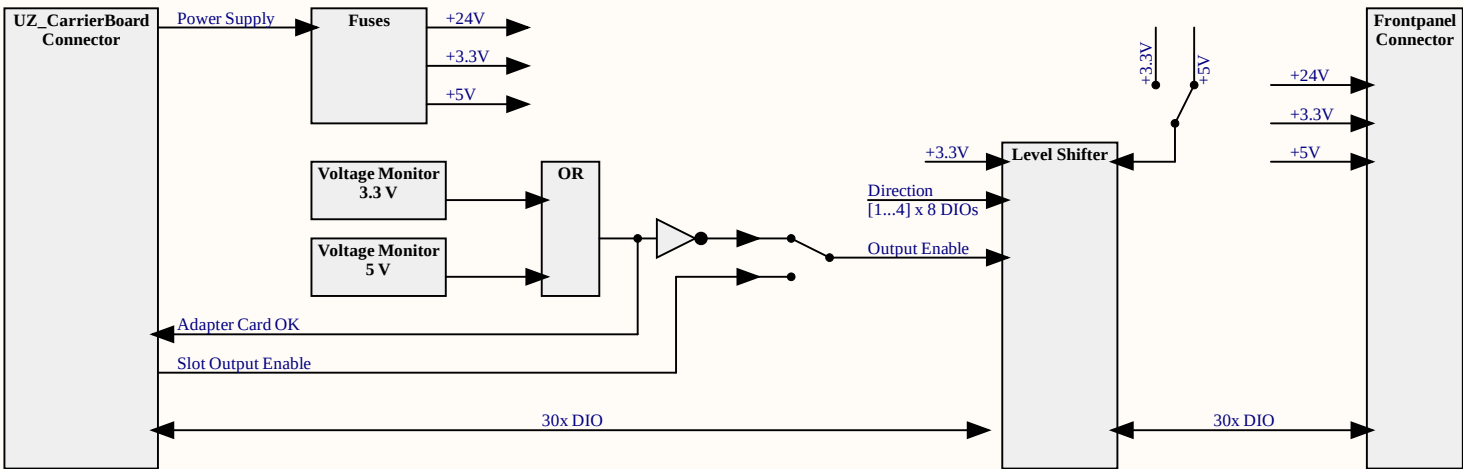


Top Sheet

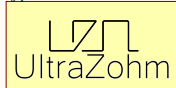


Block Diagram

- Board Features:
- configurable output voltage level (3.3V or 5V)
 - voltage monitor ICs supervise 3.3V and 5V supply rail
 - output enable of level shifters can be controlled either by the supervisors status or via UZ_CarrierBoard signal
 - I²C-Port Expander reads selected output voltage level and signal direction per group of 8 channels
 - RGB-LED at front panel visualizes selected voltage and error status



Chose size of UZ logo
by choosing footprint
type



LOGO1

INFO1

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

INFO2

ZC_GGG
ZC_PPP
ProjectRevision
ZC_vv

ZC Serialnumber

Title TopSheet.SchDoc

Revision: 03

Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PriPCB

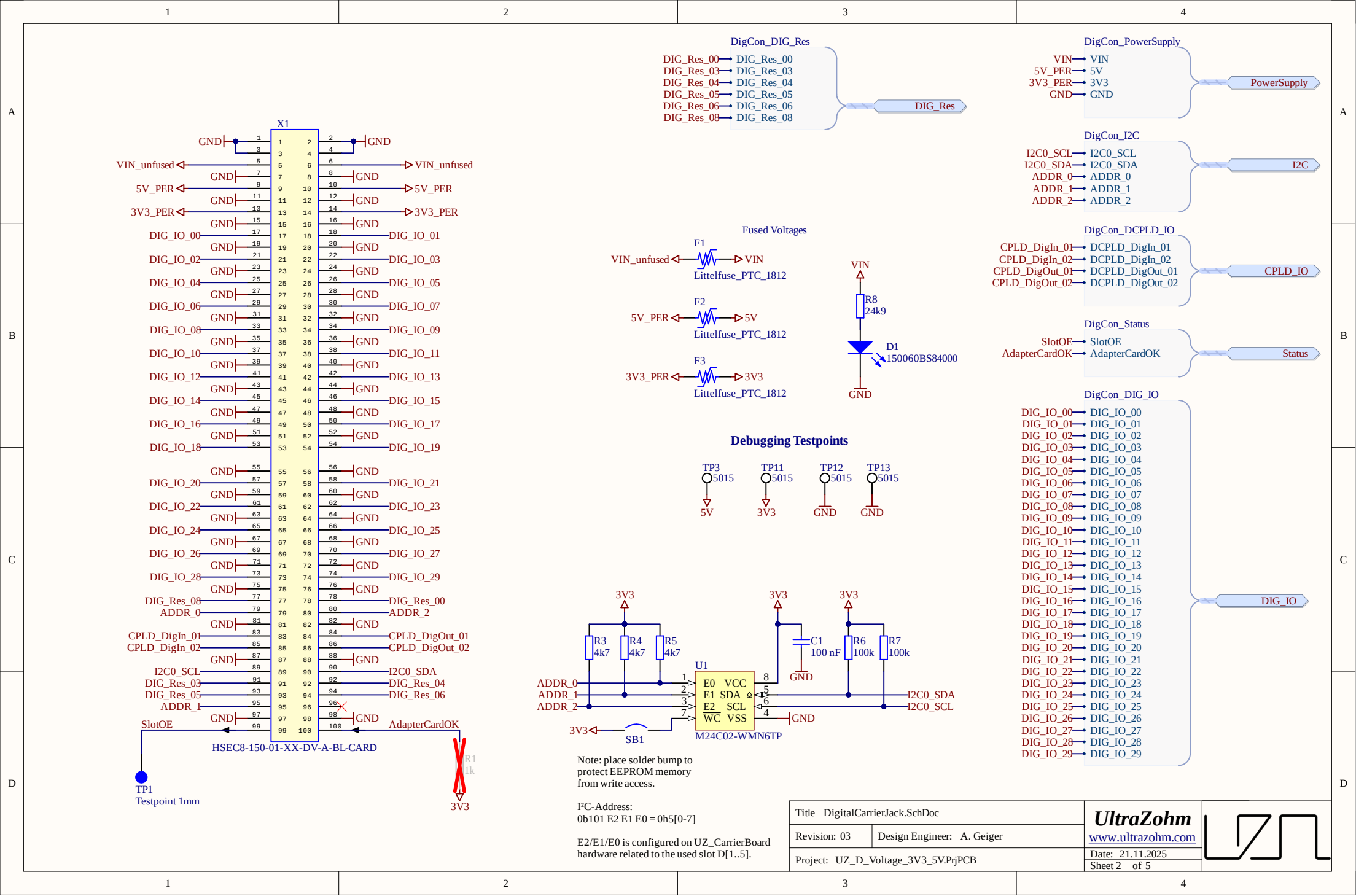
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Sheet 1 of 5





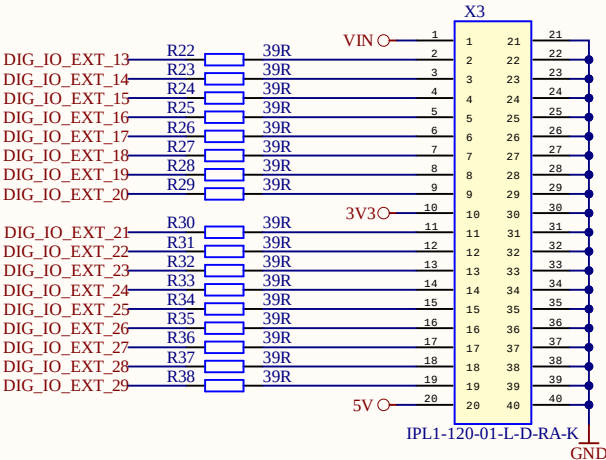
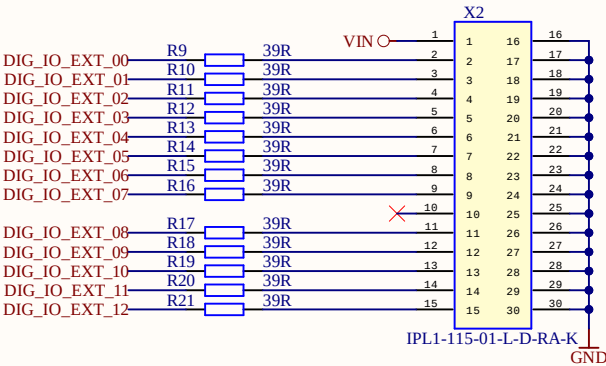
Pull-Down Resistors

Front Panel Connector

DigCon_DIG_IO_EXT

DIG_IO_EXT_00	DIG_IO_EXT_00
DIG_IO_EXT_01	DIG_IO_EXT_01
DIG_IO_EXT_02	DIG_IO_EXT_02
DIG_IO_EXT_03	DIG_IO_EXT_03
DIG_IO_EXT_04	DIG_IO_EXT_04
DIG_IO_EXT_05	DIG_IO_EXT_05
DIG_IO_EXT_06	DIG_IO_EXT_06
DIG_IO_EXT_07	DIG_IO_EXT_07
DIG_IO_EXT_08	DIG_IO_EXT_08
DIG_IO_EXT_09	DIG_IO_EXT_09
DIG_IO_EXT_10	DIG_IO_EXT_10
DIG_IO_EXT_11	DIG_IO_EXT_11
DIG_IO_EXT_12	DIG_IO_EXT_12
DIG_IO_EXT_13	DIG_IO_EXT_13
DIG_IO_EXT_14	DIG_IO_EXT_14
DIG_IO_EXT_15	DIG_IO_EXT_15
DIG_IO_EXT_16	DIG_IO_EXT_16
DIG_IO_EXT_17	DIG_IO_EXT_17
DIG_IO_EXT_18	DIG_IO_EXT_18
DIG_IO_EXT_19	DIG_IO_EXT_19
DIG_IO_EXT_20	DIG_IO_EXT_20
DIG_IO_EXT_21	DIG_IO_EXT_21
DIG_IO_EXT_22	DIG_IO_EXT_22
DIG_IO_EXT_23	DIG_IO_EXT_23
DIG_IO_EXT_24	DIG_IO_EXT_24
DIG_IO_EXT_25	DIG_IO_EXT_25
DIG_IO_EXT_26	DIG_IO_EXT_26
DIG_IO_EXT_27	DIG_IO_EXT_27
DIG_IO_EXT_28	DIG_IO_EXT_28
DIG_IO_EXT_29	DIG_IO_EXT_29

DIG_IO_EXT_00	R42	100k	GND
DIG_IO_EXT_01	R73	100k	GND
DIG_IO_EXT_02	R74	100k	GND
DIG_IO_EXT_03	R85	100k	GND
DIG_IO_EXT_04	R86	100k	GND
DIG_IO_EXT_05	R87	100k	GND
DIG_IO_EXT_06	R88	100k	GND
DIG_IO_EXT_07	R89	100k	GND
DIG_IO_EXT_08	R90	100k	GND
DIG_IO_EXT_09	R91	100k	GND
DIG_IO_EXT_10	R92	100k	GND
DIG_IO_EXT_11	R93	100k	GND
DIG_IO_EXT_12	R94	100k	GND
DIG_IO_EXT_13	R95	100k	GND
DIG_IO_EXT_14	R96	100k	GND
DIG_IO_EXT_15	R97	100k	GND
DIG_IO_EXT_16	R98	100k	GND
DIG_IO_EXT_17	R99	100k	GND
DIG_IO_EXT_18	R100	100k	GND
DIG_IO_EXT_19	R101	100k	GND
DIG_IO_EXT_20	R102	100k	GND
DIG_IO_EXT_21	R103	100k	GND
DIG_IO_EXT_22	R104	100k	GND
DIG_IO_EXT_23	R105	100k	GND
DIG_IO_EXT_24	R106	100k	GND
DIG_IO_EXT_25	R107	100k	GND
DIG_IO_EXT_26	R108	100k	GND
DIG_IO_EXT_27	R109	100k	GND
DIG_IO_EXT_28	R110	100k	GND
DIG_IO_EXT_29	R111	100k	GND



Title Front_Connector.SchDoc

Revision: 03

Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PrjPCB

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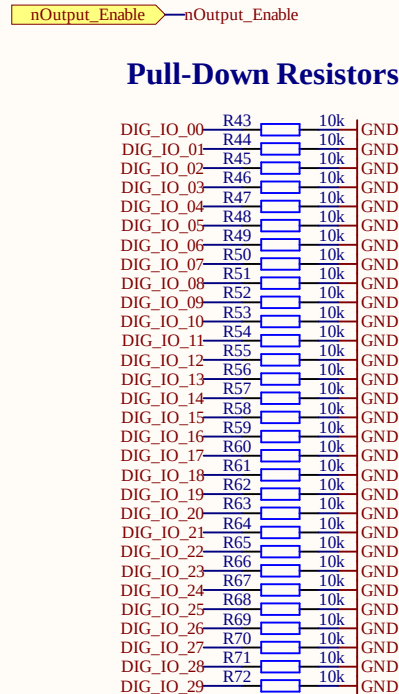
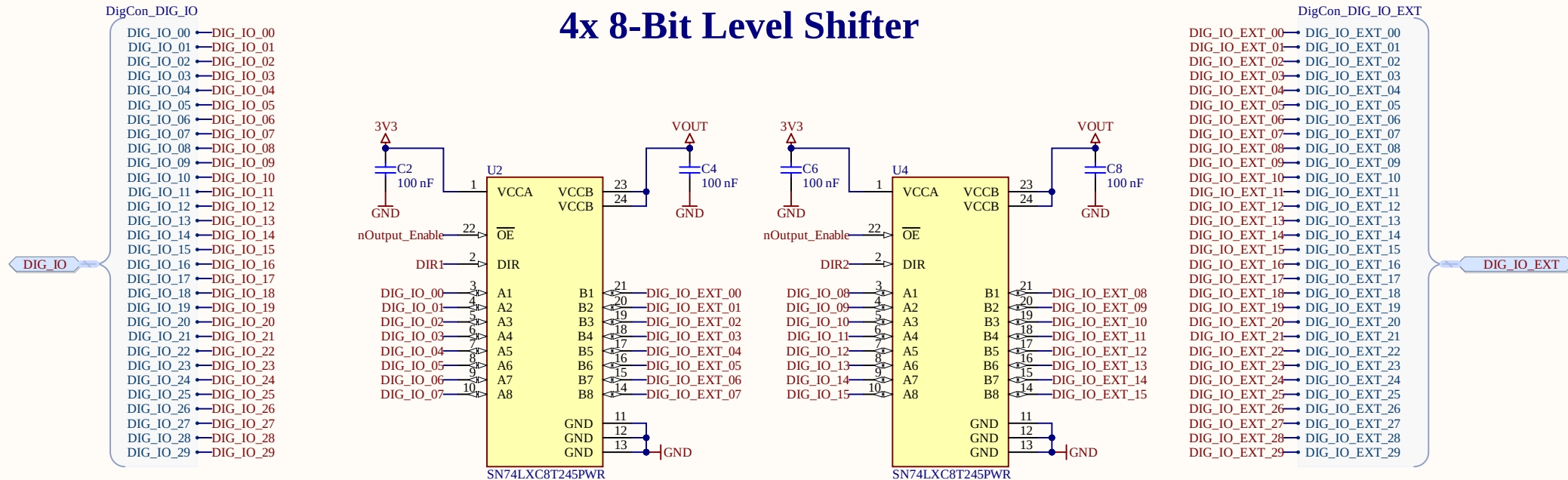
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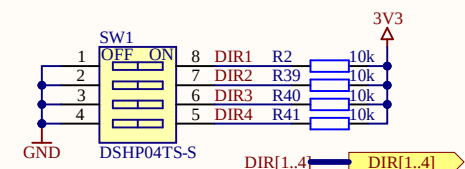
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4x 8-Bit Level Shifter



Direction per Level Shifter

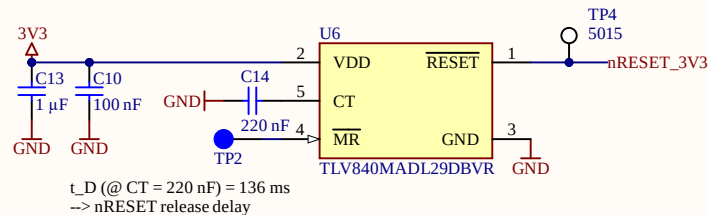


CONTROL INPUTS		Port Status		OPERATION
OE	DIR	A PORT	B PORT	
L	L	Output (H-Z)	Input (H-Z)	B data to A bus
L	H	Input (H-Z)	Output (Enabled)	A data to B bus
H	X	Input (H-Z)	Input (H-Z)	Isolation

(1) Input circuits of the data I/Os are always active and should be kept at a valid logic level.

DIP-Switch	DIR	Signal Direction
OFF	HIGH	Output
ON	LOW	Input

3.3 V Voltage Monitoring



3V3 Rail	/RESET
0 ... 0,7 V	"LOW" (follows 3V3 rail voltage)
0,7 ... 2,9 V	LOW
2,9 ... 3,3 V	HIGH

Voltage Monitoring Functionality

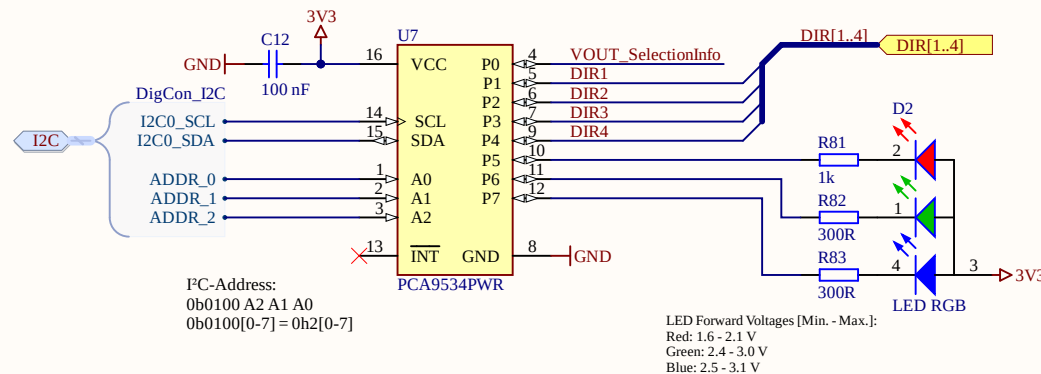
8.4 Device Functional Modes

Table 8-1 summarizes the various functional modes of the device. Logic high is represented by "H" and logic low is represented by "L".

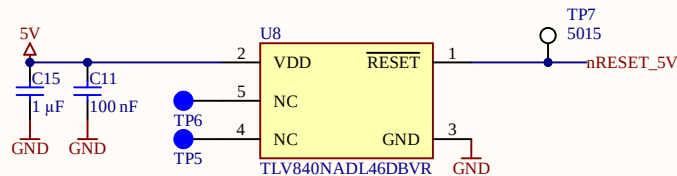
Table 8-1. Truth Table

VDD	MR	RESET	RESET
$VDD < V_{POR}$	Ignored	Undefined	Undefined
$V_{POR} < V_{DD} < V_{IT}$	Ignored	H	L
$VDD \geq V_{IT}$	L	H	L
$VDD \geq V_{IT}$	H	L	H
$VDD \geq V_{IT}$	Floating	L	H

I²C-Port Expander



5 V Voltage Monitoring



5V Rail	/RESET
0 ... 0,7 V	"LOW" (follows 3V3 rail voltage)
0,7 ... 4,6 V	LOW
4,6 ... 5 V	HIGH

8.4.1 Normal Operation ($V_{DD} > VPOR$)

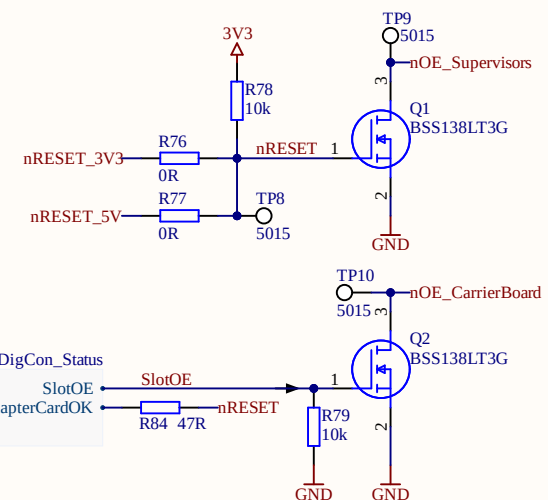
When VDD is greater than VPOR, the reset signal is determined by the voltage on the VDD pin with respect to the trip point (V_{IT} .)

- $\overline{MR}$ high: the reset signal corresponds to VDD with respect to the threshold voltage.
- $\overline{MR}$ low: in this mode, the reset is asserted regardless of the threshold voltage.

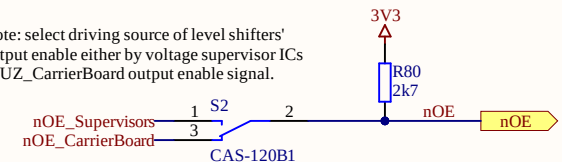
8.4.2 Below Power-On-Reset ($V_{DD} < V_{POR}$)

When the voltage on VDD is lower than V_{POR} , the device does not have enough bias voltage to internally pull the asserted output low or high and reset voltage level is undefined.

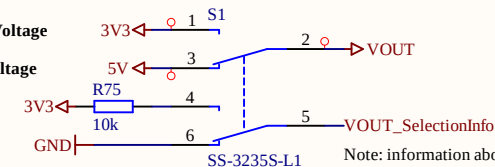
Enable Signal Logic Inversion



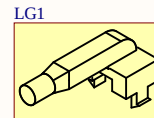
Note: select driving source of level shifters' output enable either by voltage supervisor ICs or UZ_CarrierBoard output enable signal.



Level Shifter - Output Voltage Selection



Note: information about selected output voltage
HIGH --> VOUT = 3.3 V
LOW --> VOUT = 5 V



Title Output_Voltage_Selection_and_Enable.SchDoc	
Revision: 03	Design Engineer: A. Geiger
Project: UZ_D_Voltage_3V3_5V.PrjPCB	

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